

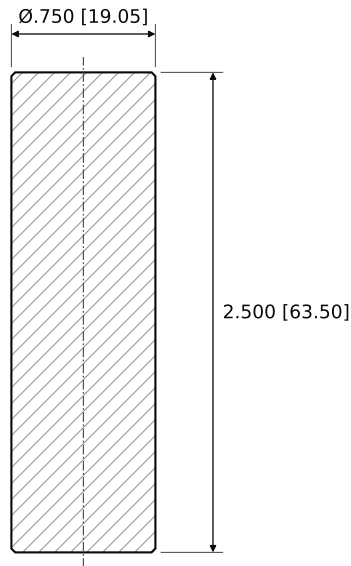
rePowder first-run charges: parts for the prototyping lab

Sections through the axis, hatched = material. Drawn from the same profiles as the STEP files · issue #222 · 2026-09-24

P1 Solid slug

6063-T52 · qty 2 (E2)

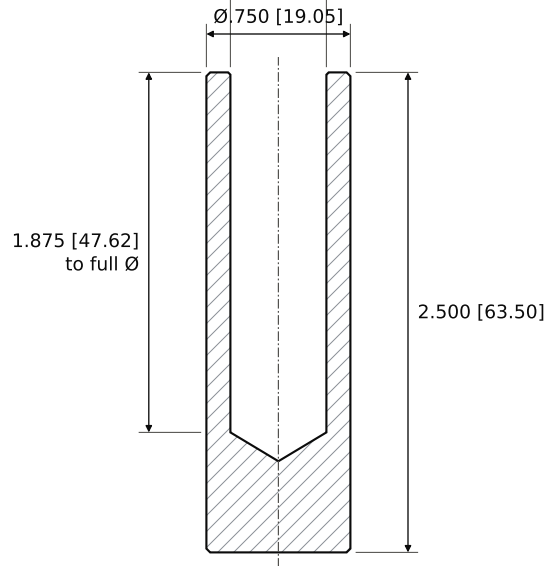
OD as received · .02 × 45° both ends



P2 Standard cup

6063-T52 · qty 10 (E1, E3-E6)

118° drill point OK · rim .012 × 45°
Ø.500 [12.70] ream

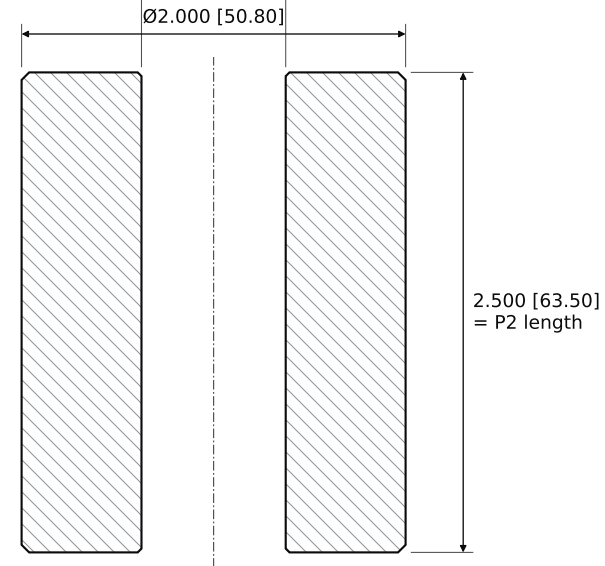


F1 Press support sleeve

mild steel (1018, 12L14) · qty 1 · E4 only

.02 × 45° both ends, bore lead-ins

Ø = measured P2 OD + .001-.002 (slip)



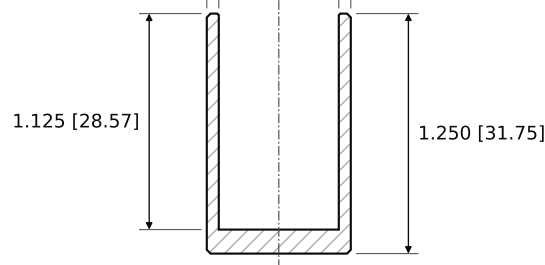
P4 Thin-wall cup

6063-T52 · qty 2 (E7, later)

1/16" wall · .125 [3.17] floor

Ø.625 [15.88] flat bottom

Ø.750 [19.05]

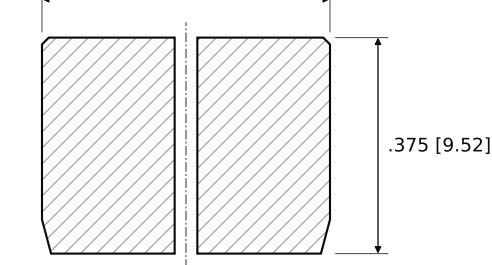


P3 Plug for P2 (3:1)

6063-T52 · qty 10 · one per cup

Ø1 mm [.040, #60] vent THRU · 15° lead-in .06 long

Ø = measured P2 bore + .0005-.001

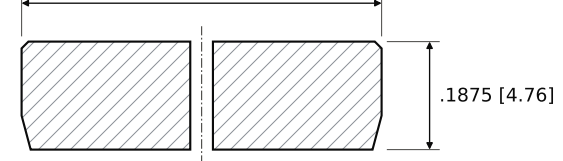


P5 Plug for P4 (3:1)

6063-T52 · qty 2

Ø1 mm vent THRU · same lead-in as P3

Ø = measured P4 bore + .0005



NOTES

- 1 P1-P5 from the 3/4" 6063-T52 bar on hand (McMaster 1640T16). Inches [mm].
- 2 Outer edges .02 [0.5] × 45°, deburr bores. No marker ink, scribing or stamping.
- 3 Make each cup first, MEASURE its bore, then turn that cup's plug to +.0005-.001" over it. Never more than .001". P2's 1/8" wall is at ~100 MPa hoop stress there (6063-T52 yield ≥ 110 MPa). Bag each cup with its own plug.
- 4 Every plug is vented: gas sealed in with powder can eject melt (#104, #134), and the chamber is pumped down before melting.

- 5 Slug OD ≤ measured rod-to-wall gap – 0.7 mm (Al grows ~1.4 % by 600 °C, graphite ~0.3 %). A max-tolerance bar (Ø.764) needs a 20.1 mm gap; otherwise skim the ODs.
 - 6 After machining: degrease in IPA, dry, weigh each part to 0.01 g, bag, label the bag.
 - 7 F1 (E4 only): stand the filled cup in F1 on a flat plate, press the plug flush. F1 is the cup's length, so the platen bottoms out at flush. Push out with a Ø5/8" drift.
 - 8 Crucible geometry is inferred, not measured: check Ø52 bore / Ø12 rod before cutting.
- Parts 1:1 printed on Letter at 100 %, plugs 3:1 · STEP: atomizer-charge/cad/step/